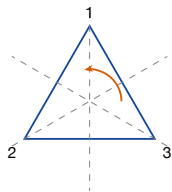


Groups

Every group is the symmetries of something

A self-contained handout for Office Hour One of *Office Hours with a Geometric Group Theorist*



3 reflections (dashed axes)

rotations by 0° , 120° , 240°

$3 + 3 = 6$ symmetries $= |S_3|$

Where we are going

A **symmetry** of an object is a move that leaves it looking exactly as it did. To understand an object, find its symmetries. This handout follows the first chapter of the book and is written so that it can be read without it.

- 1 **A puzzle.** How many ways can you turn a cube, and how do the turns combine?
- 2 **What a group is**, and why every collection of symmetries is one.
- 3 **A tour of examples:** polygons (D_n), permutations (S_n), clocks ($\mathbb{Z}/n\mathbb{Z}$), the integers, the grid, matrices, free groups. Each comes with the picture it is the symmetries of.
- 4 **Maps between groups:** homomorphisms, kernels, normal subgroups, quotients, the first isomorphism theorem.
- 5 **Writing a group down:** generators, relations, and the Cayley graph as the picture of a presentation.

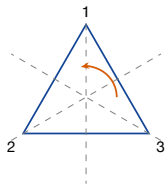
Big picture

The one sentence. “Group” and “the symmetries of something” are the same idea seen from two sides. Symmetries can be *combined* (do one, then another) and *undone*; a group is exactly a set with those two features. Every example here is both a group and a set of symmetries.

Try it

Boxes like this one contain things to check by hand and hints for the chapter’s thirteen exercises. Green boxes give the reason behind a claim; red boxes flag the usual misreadings.

A warm-up: the triangle



3 reflections (dashed axes)
rotations by 0° , 120° , 240°
 $3 + 3 = 6$ symmetries $= |S_3|$

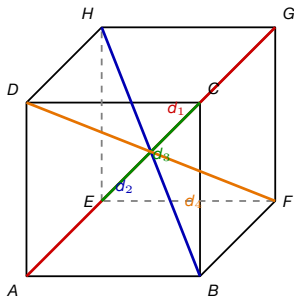
The equilateral triangle has six symmetries: three rotations (by 0° , 120° , 240°) and three reflections (in the dashed axes). Number the corners 1, 2, 3. Each symmetry moves corners to corners, so it *rearranges* $\{1, 2, 3\}$, and the six symmetries give the six possible rearrangements, each exactly once.

Two things you can always do with symmetries:

- **combine** them: do one, then another; the result is again a symmetry;
- **undo** them: every symmetry has a reverse move.

These two features are all that “group” will mean.

A puzzle: how many ways can you turn a cube?



Corners A–H; long diagonals $d_1 = AG$, $d_2 = BH$,
 $d_3 = CE$, $d_4 = DF$.

A **symmetry of the cube** is what you get by picking the cube up, turning it, and putting it back in exactly the same place. Mirror images are not allowed: you cannot do those with your hands.

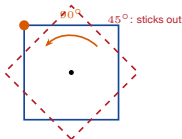
- Skewer the cube through the centres of two opposite faces and turn by a multiple of 90° .
- Or skewer it through the midpoints of two opposite edges, or through two opposite corners.
- **Counting:** corner A can be sent to any of the 8 corners, and once you have chosen, the cube can still spin about that corner in 3 ways. So there are $8 \times 3 = \mathbf{24}$ symmetries.

Watch out

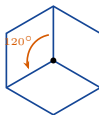
Counting is not understanding. What is a good way to *list* the 24? And if we do one turn and then another, we get a third turn: where is it on the list? The next three slides answer this.

Why the angles are 90° , 120° , 180°

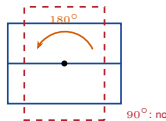
Why can you turn by 90° about a face axis but not by 45° ? Look straight down the axis you are turning about. The cube's outline becomes a flat shape, and a turn is a symmetry of the cube exactly when it is a symmetry of that outline.



looking down a face axis:
the outline is a square, 4-fold



looking down a long diagonal:
the outline is a regular hexagon, 3-fold



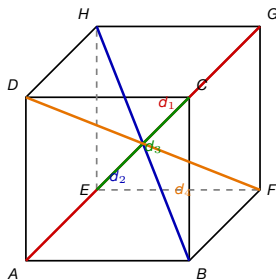
looking down an edge axis:
the outline is a $1 \times \sqrt{2}$ rectangle, 2-fold

Down a **face axis** the outline is a square: it returns to itself after 90° , and after 45° its corners stick out (red dashed). Down a **long diagonal** the outline is a regular hexagon centred on the near corner, whose three edges make 120° angles. Down an **edge axis** the outline is a rectangle with sides 1 and $\sqrt{2}$, not a square, so only the half turn works.

Why?

Count again: 1 identity + $3 \cdot 3$ (face axes) + $4 \cdot 2$ (long diagonals) + $6 \cdot 1$ (edge axes) = 24.

The idea that solves the puzzle: four long diagonals



Draw the four long diagonals (corner to opposite corner through the centre). Two facts:

- 1 Every turn moves long diagonals to long diagonals, so it **permutes** d_1, d_2, d_3, d_4 .
- 2 Every permutation of the four diagonals comes from **exactly one** turn.

Why?

Fact 2: swapping $d_1 = AG$ with $d_2 = BH$ is the half turn about the axis through the midpoints of the edges AB and GH , which swaps $A \leftrightarrow B$ and $G \leftrightarrow H$. So every swap of two diagonals is a turn, and any permutation is a product of swaps. Both sides have 24 elements, so no two turns give the same permutation.

Big picture

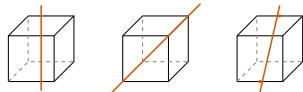
A permutation is a bijection, so “do one turn, then another” is composition of functions: mechanical. The 24 moves have become the **symmetric group** S_4 .

The dictionary: each kind of turn is a kind of permutation

rotation of the cube	how many	permutation of $d_1, \dots, d_4$	how many
do nothing	1	identity $()$	1
$\pm 90^\circ$ about a face axis	6	4-cycle, e.g. $(d_1 d_2 d_3 d_4)$	6
180° about a face axis	3	two swaps, e.g. $(d_1 d_2)(d_3 d_4)$	3
$\pm 120^\circ$ about a long diagonal	8	3-cycle, e.g. $(d_2 d_3 d_4)$	8
180° about an edge axis	6	one swap, e.g. $(d_1 d_2)$	6
total	24		24

Why?

Why a 120° turn about d_1 is a 3-cycle: the axis is d_1 itself, so d_1 stays. The three corners next to A are B, D, E and the turn cycles them; d_2, d_4, d_3 start at those corners, so they are cycled. (Cycle notation is explained on the S_n slide.)



face axis: 3 of them, turns by 90° or 180° ; vertex axis: 4, turns by 120° ; edge axis: 6, turns by 180°

The three kinds of axis. Face axes: 3, quarter turns. Corner axes: 4, third turns.

Edge axes: 6, half turns.

Definition of a group

Definition

A **multiplication** on a set G is a function $G \times G \rightarrow G$; the image of (g, h) is written gh . A **group** is a set G with a multiplication such that

- (i) **identity**: there is $1 \in G$ with $1g = g1 = g$ for all g ;
- (ii) **inverses**: for each g there is g^{-1} with $gg^{-1} = g^{-1}g = 1$;
- (iii) **associativity**: $(fg)h = f(gh)$ for all f, g, h .

Why?

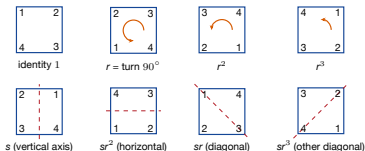
For the symmetries of the cube define gh as “do h , then do g ”, like the composition $g \circ h$. The identity is “leave it alone”; g^{-1} is “undo g ”; associativity holds because composing functions is associative. So the symmetries of anything form a group. The symbol 1 is a name, not the number one: in $\mathbb{Z}$ the identity is 0 .

Watch out

Nothing says $gh = hg$. Groups where it always holds are **abelian**. The cube's group is not: a quarter turn about a vertical axis followed by a half turn about an edge axis differs from the same two moves in the other order.

Generators. A subset S *generates* G if every element of G is a product of elements of S and their inverses.

The dihedral group D_n : symmetries of a regular polygon



The eight symmetries of a square, D_4 : four rotations and four reflections. Corners are numbered so you can see where each goes.

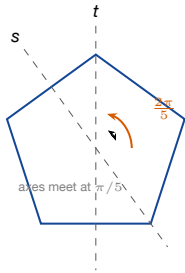
D_n is the set of symmetries of a regular n -gon in the plane, now including reflections (you can flip a flat polygon over). It has n rotations and n reflections, so $|D_n| = 2n$.

- s : reflection in a line through the centre and a vertex.
- t : reflection in the line rotated from it by π/n .
- s and t **generate** D_n : every symmetry is a product of s 's and t 's. A rotation by m clicks is $(st)^m$; each reflection is a rotation followed by s .
- $s^2 = 1$: reflecting twice puts everything back, so s is its own inverse.

Watch out

Some books call this group D_{2n} , naming it by its size. This book names it by the polygon.

Why st is one click



Why?

Reflecting in two lines that meet at angle θ is the same as rotating by 2θ about their intersection. Here $\theta = \pi/n$, so st is the rotation by $2\pi/n$.

Consequences: $(st)^m$ is m clicks, so all n rotations are $(st)^m$ for $m = 0, \dots, n-1$; every reflection is (rotation)· s ; hence s and t generate everything. In D_5 , $(st)^5 = 1$.

Try it

Take a square and number its corners. Compose two of the pictures on the previous slide: do r then s , and do s then r . You land on two *different* reflections. That is $sr \neq rs$: D_4 is not abelian.

The symmetric group S_n : all permutations of $\{1, \dots, n\}$

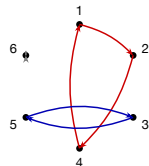
S_n is the set of permutations (bijections) of $\{1, \dots, n\}$, with multiplication given by composition. It is the symmetry group of n loose points: nothing to respect but the points, so every rearrangement counts, and $|S_n| = n!$.

Cycle notation. $(i_1 i_2 \dots i_k)$ sends each listed number to the next, the last back to the first, and fixes everything else: $(1\ 2\ 4)$ sends $1 \rightarrow 2, 2 \rightarrow 4, 4 \rightarrow 1$. Every permutation is a product of *disjoint* cycles, e.g. $(1\ 2\ 4)(3\ 5)$ in S_6 .

Multiplying is composing, right factor first. For $(2\ 6) \cdot (1\ 2\ 4)(3\ 5)$:

x	1	2	3	4	5	6
after $(1\ 2\ 4)(3\ 5)$	2	4	5	1	3	6
after $(2\ 6)$	6	4	5	1	3	2

Reading the last row: $1 \rightarrow 6 \rightarrow 2 \rightarrow 4 \rightarrow 1$ and $3 \leftrightarrow 5$, so the product is $(1\ 6\ 2\ 4)(3\ 5)$.

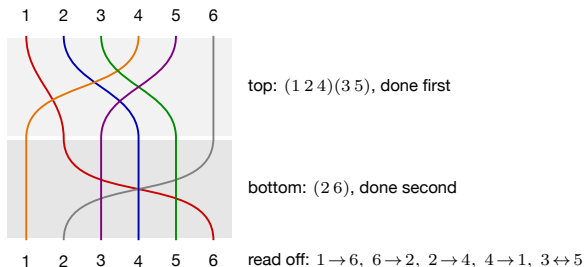


$(1\ 2\ 4)(3\ 5)$ as arrows:
red 3-cycle, blue 2-cycle, 6 fixed

Watch out

Apply the left factor first and you get $(1\ 2\ 4\ 6)(3\ 5)$, a different element. Order matters; the convention is “right factor first”, matching $gh = “h, \text{ then } g”$.

Multiplying by stacking diagrams



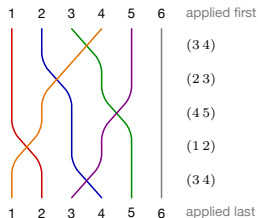
Draw a permutation as strands from a top row to a bottom row: strand i runs from position i to position $\sigma(i)$.

To multiply, **stack**: the first-applied permutation on top, the second underneath. Follow each colour through both blocks. The red strand starts at 1, reaches 2 in the top block, then 6 in the bottom block: $1 \mapsto 6$. Reading all six strands gives $(1\ 6\ 2\ 4)(3\ 5)$, the same answer as the table.

Why?

Why is the first-applied permutation on top? Strands are read top to bottom, and σ is applied first. This matches “do h , then g ” = gh .

Transpositions: neighbour swaps generate S_n



The strand diagram of $(1\ 2\ 4)(3\ 5)$; each crossing is a swap of two neighbouring positions.

A **transposition** is a 2-cycle such as $(3\ 5)$: it swaps two things and fixes the rest. Important fact: S_n is generated by the *adjacent* transpositions $(i\ i+1)$, $1 \leq i \leq n - 1$. (We used this on the cube: every permutation of the diagonals is a product of swaps.)

In a strand diagram, each crossing swaps two neighbouring positions, so it is an adjacent transposition. Read the crossings top to bottom and write them right to left (top crossing = applied first = rightmost factor):

$$(1\ 2\ 4)(3\ 5) = (3\ 4)(1\ 2)(4\ 5)(2\ 3)(3\ 4).$$

Why?

Multiplying by stacking (previous slide) means the diagram of a product is the stack of the diagrams of its factors. The five crossings, each at its own height, are five one-crossing diagrams stacked: five adjacent transpositions.

Exercises 1 and 2

Exercise 1, as an algorithm

Any strand diagram can be drawn with its crossings at different heights, and then it *is* a product of adjacent transpositions. Equivalently, bubble sort: start from the arrangement 2, 4, 5, 1, 3, 6 that the permutation $(1\ 2\ 4)(3\ 5)$ produces and sort it back into order using only swaps of neighbours. Each swap is an adjacent transposition, and the list of swaps is a factorisation of the inverse permutation; read it in reverse to get the permutation itself.

Try it

Sort 2, 4, 5, 1, 3, 6 by hand. The first pass swaps positions 3–4, then 4–5; keep going until sorted and count the swaps.

Exercise 2

Which other S_n are the symmetries of a shape? The regular tetrahedron has 4 corners; its rotations form a group of 12 elements, and rotations together with reflections give all $4! = 24$ permutations of the corners, i.e. S_4 . In general the regular *simplex* with n corners (segment, triangle, tetrahedron, ...) has full symmetry group S_n , because any rearrangement of its corners extends to a symmetry. Higher cubes do *not* give symmetric groups: the 4-dimensional cube has 192 rotations, while $5! = 120$ and $6! = 720$.

$\mathbb{Z}/n\mathbb{Z}$: add, then wrap around

Definition

For an integer $n > 1$, $\mathbb{Z}/n\mathbb{Z} = \{0, 1, \dots, n-1\}$ with the multiplication “add a and b , and subtract n if the result is n or more”. The identity is 0; the inverse of $m \neq 0$ is $n - m$.

Three familiar cases: $n = 12$ is clock arithmetic (9 o'clock plus 4 hours is 1 o'clock); $n = 10$ is “only keep the last digit”; $n = 2$ is light-switch arithmetic, where 1 means flip and flip plus flip is nothing.

Why?

Associativity: the definition is by cases, but $(a + b) + c$ and $a + (b + c)$ are both the remainder of the ordinary integer $a + b + c$ on division by n , and taking remainders is consistent with integer addition.

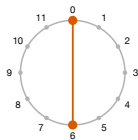


$$9 + 4 = 13 \equiv 1 \pmod{12}$$

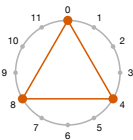
“13” is not an element;
subtract $n = 12$ to land in $\{0, \dots, 11\}$.

Subgroups. A *subgroup* of G is a subset containing 1 that is closed under products and inverses; it is a group in its own right. “Proper” means not the whole group.

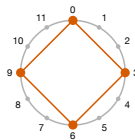
Generators of $\mathbb{Z}/n\mathbb{Z}$ (Exercise 3), and subgroups you can see



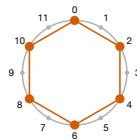
$$\{0, 6\} \cong \mathbb{Z}/2\mathbb{Z}$$



$$\{0, 4, 8\} \cong \mathbb{Z}/3\mathbb{Z}$$



$$\{0, 3, 6, 9\} \cong \mathbb{Z}/4\mathbb{Z}$$



$$\{0, 2, \dots, 10\} \cong \mathbb{Z}/6\mathbb{Z}$$

Exercise 3

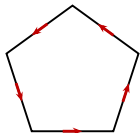
If $\gcd(m, n) = 1$ then $\{m\}$ generates $\mathbb{Z}/n\mathbb{Z}$. Why: there are integers u, v with $um + vn = 1$ (Bézout), so $u \cdot m \equiv 1$, and from 1 you reach everything. Conversely if $d = \gcd(m, n) > 1$, every multiple of m is a multiple of d on the clock, so 1 is never reached.

Try it

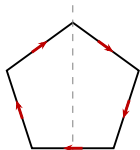
On the 12-clock start at 0 and keep adding 5: 0, 5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7, every hour visited, because $\gcd(5, 12) = 1$. Keep adding 4 instead: only 0, 4, 8, the triangle above; $\gcd(4, 12) = 4$.

Each highlighted set is closed under “add and wrap”, so each is a subgroup, and each is a regular polygon inscribed in the clock.

What is $\mathbb{Z}/n\mathbb{Z}$ the symmetries of? Decorate the polygon



rotate: arrows still agree



reflect: every arrow flips

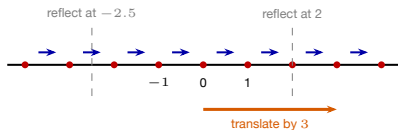
Using the clock, we might guess that $\mathbb{Z}/n\mathbb{Z}$ is the symmetries of a regular n -gon. Not quite: it is the **rotations**, a proper subgroup of D_n , because the reflections are missing.

The trick: draw a small arrow on every edge and insist that a symmetry keeps the arrows pointing the same way round. Reflections reverse the sense of travel, so they are no longer symmetries; the n rotations remain, and they compose exactly like $\mathbb{Z}/n\mathbb{Z}$: turn j clicks, then k clicks, is $j + k$ clicks, wrapped.

Big picture

This decorated polygon is the **Cayley graph** of $\mathbb{Z}/n\mathbb{Z}$: one dot per element, one arrow per “add 1”. The same recipe builds an object for every group, and Office Hour 2 proves the group is exactly its symmetry group.

$\mathbb{Z}$: the integers and the arrowed line

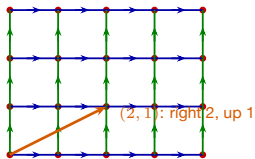


Many first courses skip infinite groups. This is where the fun is, and where pictures earn their keep: you cannot write an infinite multiplication table, but you can draw the group's object.

$\mathbb{Z}$ under addition: identity 0 , inverse of n is $-n$. Generated by $\{1\}$, and also by $\{2, 3\}$ since $3 - 2 = 1$; by $\{2\}$ alone you get only the even numbers, a proper subgroup.

Mark the integer points on a line. Its symmetries are translations by an integer and reflections about integer or half-integer points. $\mathbb{Z}$ is the translations: n acts as "slide by n ", and slide m then slide n is slide $m + n$. Draw arrows on the segments and the reflections disappear, so $\mathbb{Z}$ becomes the *full* symmetry group.

$\mathbb{Z}^2$: the integer grid



blue = horizontal edges, green = vertical edges

$\mathbb{Z}^2$ is the set of pairs (m, n) of integers, added coordinate by coordinate. Identity $(0, 0)$; generated by $(1, 0)$ and $(0, 1)$, since $(m, n) = m(1, 0) + n(0, 1)$.

Colour the horizontal edges blue and the vertical edges green, add arrows, and demand that a symmetry preserves points, colours and arrows. A quarter turn would swap blue and green, a reflection would flip arrows, so only translations by lattice vectors survive: the symmetries are exactly $\mathbb{Z}^2$.

Watch out

$\mathbb{Z}^2$ is abelian: $(m, n) + (m', n') = (m', n') + (m, n)$. The free group two slides on also has two generators but is *not* abelian. Two generators, two very different groups.

Numbers under multiplication, and matrix groups

Multiplicative groups. $\mathbb{R}^\times$ and $\mathbb{C}^\times$ are the nonzero real and complex numbers under ordinary multiplication, identity 1. Zero is left out because it has no inverse.

Why?

Not finitely generated: finitely many numbers $r_1, \dots, r_k$ produce only the products $r_1^{e_1} \cdots r_k^{e_k}$ with integer exponents, countably many. But $\mathbb{R}^\times$ is uncountable. Every group so far had a finite generating set; these do not.

Matrix groups. $GL(n, \mathbb{R})$, the *general linear group*, is the set of real $n \times n$ matrices with nonzero determinant. It is a group because:

- closed: $\det(MN) = \det M \cdot \det N \neq 0$;
- the identity matrix has determinant 1;
- nonzero determinant means invertible;
- matrix multiplication is associative (it is composition of linear maps).

Relatives: $SL(n, \mathbb{R})$ (determinant 1), $GL(n, \mathbb{Z})$ (integer entries, which forces $\det = \pm 1$), $SL(n, \mathbb{Z})$, and the complex versions.

Exercise 4

Over a finite field: $GL(2, \mathbb{Z}/2\mathbb{Z})$, the invertible 2×2 matrices with entries 0, 1, has 6 elements and behaves like S_3 .

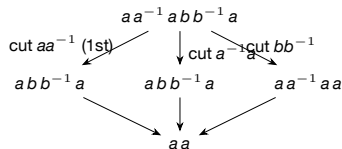
The free group F_2 : words with no rules except cancellation

Definition

A **word** in a, b is any finite string of the symbols a, b, a^{-1}, b^{-1} (the empty word included); words are multiplied by **concatenation**. A word is **reduced** if no a stands next to an a^{-1} and no b next to a b^{-1} ; deleting such a pair shortens a word, and repeating this reaches a reduced word.

F_2 is the set of reduced words with multiplication “**concatenate, then reduce**”.

- Identity: the empty word. Inverse: reverse the word and invert each letter, $(aba^{-1}b)^{-1} = b^{-1}ab^{-1}a^{-1}$ (socks and shoes).
- F_2 is generated by $\{a, b\}$. Not abelian: ab and ba are different reduced words, hence different elements.

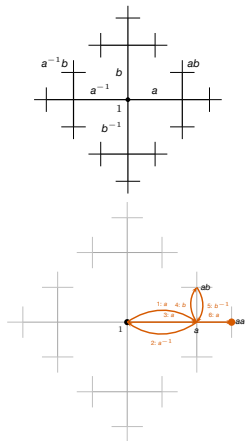


Three possible first steps, one final answer.

Exercises 5 and 6

Reduction always ends at the *same* word, by induction on length: if two cancelling pairs do not overlap, cancel both in either order; if they overlap, the word contains $xx^{-1}x$ and either cancellation leaves x . This is what makes $(uv)w = u(vw)$: both are reductions of the one string uvw .

Words are walks in a tree



The walk for $aa^{-1}abb^{-1}a$: steps 1–2 and 4–5 go out and straight back; what remains is the straight path aa .

Read a word letter by letter from the centre: right for a , up for b , left for a^{-1} , down for b^{-1} . Cancelling xx^{-1} is erasing a there-and-back. A reduced word is a walk that never backtracks.

The picture continues forever, every vertex has four neighbours, and there are no loops: an infinite tree. Distinct reduced words reach distinct vertices, because in a tree there is only one non-backtracking route to each vertex. (That is another way to see the uniqueness of the reduced word.)

Big picture

This tree is the Cayley graph of F_2 , and Office Hour 2 shows that F_2 is exactly its group of symmetries preserving the letters and arrows. Compare the arrowed polygon $(\mathbb{Z}/n\mathbb{Z})$, the arrowed line $(\mathbb{Z})$ and the coloured grid $(\mathbb{Z}^2)$: the same construction, different groups.

Other free groups. The same construction on any alphabet S gives $F(S)$; its *rank* is $|S|$. F_1 is the powers of one letter, a copy of $\mathbb{Z}$. Every group turns out to be a quotient of a free group (Section 1.4).

Homomorphisms: maps that respect multiplication

Definition

A **homomorphism** from G to H is a function $f : G \rightarrow H$ with $f(ab) = f(a)f(b)$ for all a, b . A bijective homomorphism is an **isomorphism**; isomorphic groups are the same group with the elements renamed (e.g. $\mathbb{Z} \rightarrow F_1, n \mapsto a^n$).

$$\begin{array}{ccc} (a, b) \in G \times G & \xrightarrow{f \text{ on each}} & (f(a), f(b)) \in H \times H \\ \downarrow \text{multiply in } G & & \downarrow \text{multiply in } H \\ ab \in G & \xrightarrow{f} & f(a)f(b) = f(ab) \in H \end{array}$$

Multiply then map, or map then multiply: same answer.

Two consequences: $f(1) = 1$ (since $f(1) = f(1 \cdot 1) = f(1)f(1)$, then cancel) and $f(g^{-1}) = f(g)^{-1}$ (since $f(g)f(g^{-1}) = f(1) = 1$).

A homomorphism is **injective** exactly when only 1 maps to 1: if $f(g) = f(h)$ then $f(gh^{-1}) = 1$, so $gh^{-1} = 1$.

Any homomorphism becomes onto by shrinking the target to its image, so what matters is injective versus non-injective.

Injective homomorphisms: embedding one group in another

An injective homomorphism $f : G \rightarrow H$ realises G as a subgroup of H , namely its image $f(G)$.

Examples:

- $\mathbb{Z}/n\mathbb{Z} \rightarrow D_n, m \mapsto \text{rotation by } 2\pi m/n$: the clock is the rotations of the polygon.
- $\mathbb{Z}/2\mathbb{Z} \rightarrow D_n$ (nontrivial element to a reflection) and $\mathbb{Z}/2\mathbb{Z} \rightarrow S_n$ (to a product of disjoint swaps): both square to 1, so each is a copy of $\mathbb{Z}/2\mathbb{Z}$.
- $\mathbb{Z} \rightarrow \mathbb{Z}, n \mapsto mn$ for $m \neq 0$: injective, with image $m\mathbb{Z}$. So $\mathbb{Z}$ contains proper copies of itself, something no finite group can do.
- $\mathbb{Z} \rightarrow \mathbb{Z}^2, n \mapsto (na, nb)$ for $(a, b) \neq (0, 0)$: a line inside the lattice.
- $\mathbb{Z} \rightarrow F_2, n \mapsto w^n$ for a nontrivial reduced word w : the powers of a word never fully cancel.

Exercise 7

Find more: $D_3 \rightarrow S_3$ (a symmetry of the triangle permutes its 3 corners); $\mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/mk\mathbb{Z}, 1 \mapsto k$; $F_2 \rightarrow F_3$ (use two of the three letters); $\mathbb{Z}^2 \rightarrow \mathbb{Z}^3$.

Non-injective homomorphisms: forgetting on purpose (Exercise 8)

A non-injective homomorphism sends several elements to the same place, so it loses information. Because it respects multiplication, what it keeps is *consistent*: to know the property of a product, it is enough to know the property of the factors.

map	remembers only
$\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$	parity
$\det : \mathrm{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}^\times$	the determinant
$\mathbb{R}^\times \rightarrow \mathbb{Z}/2\mathbb{Z}$	the sign
$S_n \rightarrow \mathbb{Z}/2\mathbb{Z}$	even or odd number of crossings
$D_n \rightarrow \mathbb{Z}/2\mathbb{Z}$	flipped over or not
$F_2 \rightarrow \mathbb{Z}^2$	the exponent sums of a and of b

Why?

Example: the sign of a product of nonzero reals depends only on the signs of the factors. That sentence is the statement that $\mathbb{R}^\times \rightarrow \mathbb{Z}/2\mathbb{Z}$ is a homomorphism. Likewise $\det(MN) = \det M \det N$ says the determinant is one.

Watch out

For $S_n \rightarrow \mathbb{Z}/2\mathbb{Z}$ there is a hidden claim: the same permutation can be drawn with different numbers of crossings, so why is the *parity* always the same? Count inversions (pairs $i < j$ with $\sigma(i) > \sigma(j)$): each adjacent swap changes the count by exactly ± 1 .

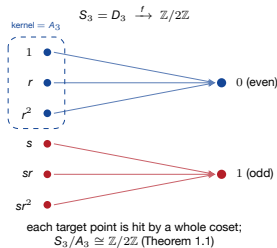
Exercise 8

More: $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ (remainder); $\mathbb{Z}^2 \rightarrow \mathbb{Z}$, $(m, n) \mapsto m - n$; $F_2 \rightarrow \mathbb{Z}/2\mathbb{Z}$ by the parity of the word length.

Normal subgroups and kernels

Definition

A **normal subgroup** of G is a subgroup N with $gng^{-1} \in N$ for every $n \in N$ and $g \in G$. The element gng^{-1} is the **conjugate** of n by g : “ n , seen from g ’s point of view” (move by g^{-1} , do n , move back). The **kernel** of a homomorphism f is the set of elements sent to the identity.



Why?

Kernels are normal: if $f(n) = 1$ then $f(gng^{-1}) = f(g) \cdot 1 \cdot f(g)^{-1} = 1$.

Kernels of the forgetful maps: $2\mathbb{Z}$; $SL(n, \mathbb{R})$; the positive reals; the alternating group A_n (even permutations); the rotations of the polygon; the commutator subgroup $[F_2, F_2]$, the words whose a -exponents and b -exponents both sum to zero.

Watch out

Not every subgroup is normal. In D_3 the rotations are normal (a conjugate of a rotation is a rotation), but $\{1, s\}$ is not: rsr^{-1} is the reflection in the axis rotated by 120° . In an abelian group $gng^{-1} = n$, so every subgroup is normal.

Quotient groups: declare every element of N trivial (Exercise 9)

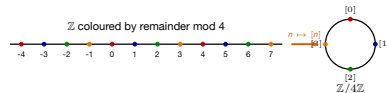
Definition

Call $g_1 \sim g_2$ if $g_1 g_2^{-1} \in N$, and write $[g]$ for the class of g . The classes are the **cosets** Ng ; the class of 1 is N . G/N is the set of classes with multiplication $[g_1][g_2] = [g_1 g_2]$.

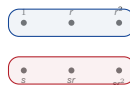
Watch out

The one computation to know. Change representatives: $(n_1 g_1)(n_2 g_2) = n_1 (g_1 n_2 g_1^{-1}) g_1 g_2$. This lies in the class of $g_1 g_2$ exactly when $g_1 n_2 g_1^{-1} \in N$. So the multiplication is well defined precisely when N is normal: the definition of “normal” is what the quotient needs.

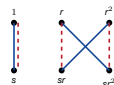
The map $g \mapsto [g]$ is a homomorphism $G \rightarrow G/N$ with kernel N . So the loop closes: **normal subgroups and surjective homomorphisms are the same thing.**



$\mathbb{Z}$ coloured by remainder mod 4: the colours are the cosets of $4\mathbb{Z}$, i.e. the elements of $\mathbb{Z}/4\mathbb{Z}$.



$N = \{1, r, r^2\}$ is normal:
left cosets gN and right cosets Ng agree



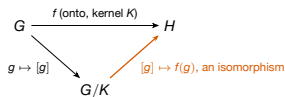
$H = \{1, s\}$ is not normal:
left cosets gH (solid) $\neq$ right cosets Hg (dashed)

For the rotations of D_3 left and right cosets agree; for $\{1, s\}$ they do not.

The First Isomorphism Theorem (Exercise 10)

Theorem 1.1

If $f : G \rightarrow H$ is a surjective homomorphism with kernel K , then $H \cong G/K$.



Why?

Define $\phi([g]) = f(g)$. Well defined: if $[g] = [g']$ then $g' = kg$ with $k \in K$, so $f(g') = f(k)f(g) = f(g)$. Homomorphism: $\phi([g][g']) = \phi([gg']) = f(gg') = f(g)f(g')$. Onto because f is. One-to-one: $\phi([g]) = 1$ means $g \in K$, i.e. $[g] = [1]$.

The examples, read as slogans:

- $\mathbb{Z}/2\mathbb{Z}$ = integers modulo the evens: parity. (This explains the notation.)
- $\text{GL}(n, \mathbb{R})/\text{SL}(n, \mathbb{R}) \cong \mathbb{R}^\times$: matrices modulo “determinant 1” is the determinant.
- $S_n/A_n \cong \mathbb{Z}/2\mathbb{Z}$: even or odd. $D_n/\text{rotations} \cong \mathbb{Z}/2\mathbb{Z}$: flipped or not.
- $F_2/[F_2, F_2] \cong \mathbb{Z}^2$: forcing $ab = ba$ in the free group produces $\mathbb{Z}^2$. This is the presentation story of the next section.

Exercise 10

$\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ by parity has kernel $\{0, 2, 4\}$, so $(\mathbb{Z}/6\mathbb{Z})/\{0, 2, 4\} \cong \mathbb{Z}/2\mathbb{Z}$. What do $\mathbb{C}^\times \rightarrow \mathbb{R}^\times, z \mapsto |z|$, and $\mathbb{Z}^2 \rightarrow \mathbb{Z}, (m, n) \mapsto m - n$, give?

How do you write a group down? (Exercise 11)

row · column:

	1	r	r^2	s	sr	sr^2
1	1	r	r^2	s	sr	sr^2
r	r	r^2	1	sr^2	s	sr
r^2	r^2	1	r	sr	sr^2	s
s	s	sr	sr^2	1	r	r^2
sr	sr	sr^2	s	r^2	1	r
sr^2	sr^2	s	sr	r	r^2	1

The multiplication table of D_3 : rotations blue, reflections red. The whole table follows from $s^2 = 1$, $r^3 = 1$, $rs = sr^2$.

A multiplication table describes a group completely but is hopeless for big or infinite groups. A generating set is more economical but not enough: F_2 and $\mathbb{Z}^2$ both have two generators, and only one is abelian.

The way out: some equations imply others (from $gh = k$ and $h = pq$ follows $gpq = k$), so a short list of **defining equations** can pin down the whole table.

Big picture

$\mathbb{Z}/n\mathbb{Z} \cong \langle a \mid a^n = 1 \rangle$: every entry follows from the one rule, e.g. $a^{n-3}a^4 = a^{n+1} = a^n \cdot a = a$.

$\mathbb{Z}^2 \cong \langle x, y \mid xy = yx \rangle$: two generators and the single rule that they commute.

Exercise 11

Using $xy = yx$ you can move every x to the left past every y , so any word becomes $x^m y^n$, and different (m, n) give different elements: that is $\mathbb{Z}^2$.

The formal definition of a presentation

Definition

A presentation is a pair (S, R) : an alphabet S and a set R of words in it. The group $\langle S \mid R \rangle$ is the free group $F(S)$ divided by the **normal closure** of R , the smallest normal subgroup containing R . Elements of S are **generators**; elements of R are **defining relators**. A **relation** $u = v$ becomes the relator uv^{-1} : $ab = ba$ becomes $aba^{-1}b^{-1}$.

$$\begin{array}{ccc} F(S) & \xrightarrow{\text{send each letter to itself}} & G \cong \langle S \mid R \rangle \\ \uparrow \text{kernel} & & \\ \langle\langle R \rangle\rangle = \text{normal closure of } R & & \end{array}$$

So the elements of $\langle S \mid R \rangle$ are reduced words, with two words equal exactly when one turns into the other by inserting or deleting conjugates of relators.

Why?

Why the normal closure? If a word r is to equal 1, then so must grg^{-1} for every g , and products and inverses of such elements as well. The set of all consequences is a normal subgroup containing R , and the smallest one is the normal closure. Dividing by a non-normal subgroup would not even give a group.

Watch out

A slip to avoid: from $aba = bab$ the relator is $aba(bab)^{-1} = aba b^{-1} a^{-1} b^{-1}$. (Socks and shoes: $(bab)^{-1} = b^{-1} a^{-1} b^{-1}$.)

Worked example: why $\langle a, b \mid aba^{-1}b^{-1} \rangle$ is $\mathbb{Z}^2$

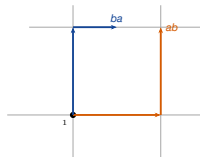
In the quotient you may insert the relator anywhere, because $wrw' = (wrw^{-1})ww'$ and wrw^{-1} is a conjugate of r . Insert it in front of ba and regroup:

$$\begin{aligned}wba w' &= w(aba^{-1}b^{-1})ba w' \\ &= wab \underbrace{(a^{-1}b^{-1}ba)}_{=1} w' = wab w'.\end{aligned}$$

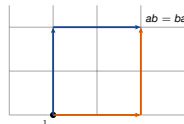
So ba has become ab : the single relator lets a and b pass through each other. Repeating sorts any word into $a^m b^n$, which is the element (m, n) of $\mathbb{Z}^2$; different (m, n) stay different because the exponent-sum map $F_2 \rightarrow \mathbb{Z}^2$ separates them.

Relations and relators say the same thing:

$\langle x, y \mid xy = yx \rangle$ and $\langle a, b \mid aba^{-1}b^{-1} \rangle$ are two descriptions of one group. The book prefers relators.



in the tree of F_2 :
 ab and ba end at different vertices

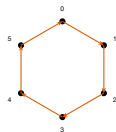


in the grid of $\mathbb{Z}^2$:
the relator $aba^{-1}b^{-1}$ is a closed square

Big picture

In the tree of F_2 the walks ab and ba end at different vertices. Imposing the relator declares the walk a, b, a^{-1}, b^{-1} a closed loop; gluing the tree along all such loops folds it into the square grid, the picture of $\mathbb{Z}^2$. That is what “quotient of a free group” looks like.

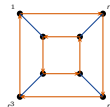
What presentations look like: Cayley graphs (Exercise 12)



$\mathbb{Z}/6\mathbb{Z} = \langle a \mid a^6 \rangle$
the relator is the loop around



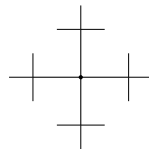
$\mathbb{Z} = \langle a \mid \rangle = F_1$
no relator, no loop



$D_4 = \langle r, s \mid r^4, s^2, (sr)^2 \rangle$
orange = r , blue = s (two-way)



$\mathbb{Z}^2 = \langle a, b \mid aba^{-1}b^{-1} \rangle$
every square is the relator



$F_2 = \langle a, b \mid \rangle$
no relators, no loops: a tree

One dot per element; from each dot an arrow per generator s , from g to gs (two-way when $s = s^{-1}$). Reading the labels along a path spells a word; a closed loop spells a word equal to 1, a relator. Office Hour 2 proves that every group is *exactly* the symmetry group of its own graph.

Exercise 12: every group is a quotient of a free group

Take $S = G$, the whole group as alphabet. The map $F(G) \rightarrow G$ sending each letter to the element it names is a homomorphism and is onto. By Theorem 1.1, $G \cong F(G)/K$ with K its kernel, so $G \cong \langle G \mid K \rangle$. Wasteful, but a presentation.

Presentations to find (Exercise 13), and the limits of presentations

Exercise 13

- $F_n = \langle a_1, \dots, a_n \mid \rangle$: no relators.
- $D_n = \langle s, t \mid s^2, t^2, (st)^n \rangle$ from the two reflections; or $\langle r, s \mid r^n, s^2, srs^{-1}r \rangle$ from a rotation r and a reflection s (the last relator says $srs^{-1} = r^{-1}$: a reflection reverses the direction of rotation).
- S_n : generators the neighbour swaps $\sigma_i = (i \ i+1)$; relators σ_i^2 , $(\sigma_i\sigma_{i+1})^3$, and $(\sigma_i\sigma_j)^2$ for $|i-j| \geq 2$. The middle one encodes $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$, the braid relation.

Writing these down is easy; proving that nothing is missing is harder.

Watch out

There is no algorithm that decides whether two finite presentations describe isomorphic groups (Adian, Rabin). Even deciding whether a given word equals 1 in $\langle S \mid R \rangle$ is impossible in general (Novikov, Boone). Presentations are useful but have limits; geometry rescues large classes of groups (Office Hour 9).

Try it

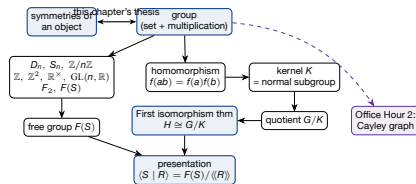
Draw the Cayley graph of D_4 with generators r and s (it is on the previous slide), start at 1, and read the word $srsr$ along the arrows. You return to 1: that is the relator $(sr)^2$.

Summary: groups lead to symmetries and symmetries lead to groups

group	multiplication	symmetries of
S_4	compose	the turns of a cube
D_n	compose	a regular n -gon
S_n	compose	n loose points
$\mathbb{Z}/n\mathbb{Z}$	add mod n	an n -gon with arrowed edges
$\mathbb{Z}, \mathbb{Z}^2$	add	arrowed line, coloured grid
$GL(n, \mathbb{R})$	matrix product	linear maps of $\mathbb{R}^n$
F_2	concatenate, reduce	the 4-valent tree
$\langle S \mid R \rangle$	words modulo relators	its Cayley graph

Big picture

Next: Office Hour 2 proves every group is the symmetry group of its Cayley graph, and turns groups themselves into geometric objects.



Glossary

symmetry	a move returning an object to its place	homomorphism	$f(ab) = f(a)f(b)$
group	set with an associative multiplication, identity, inverses	isomorphism	bijjective homomorphism; “same group”
abelian	$gh = hg$ for all g, h	kernel	elements sent to 1
generate	every element is a product of the generators and inverses	normal subgroup	$gng^{-1} \in N$ for all $g \in G, n \in N$
subgroup	subset closed under product and inverse, containing 1	conjugate	gng^{-1}
D_n	symmetries of the regular n -gon; $2n$ elements	coset, quotient G/N	classes $[g]$; $[g_1][g_2] = [g_1g_2]$
S_n	all permutations of n things; $n!$ elements	First Isomorphism Theorem	$f : G \twoheadrightarrow H$ with kernel K gives $H \cong G/K$
cycle, transposition	$(i_1 \cdots i_k)$; a 2-cycle	presentation $\langle S \mid R \rangle$	$F(S)$ modulo the normal closure of R
$\mathbb{Z}/n\mathbb{Z}$	remainders mod n under addition	relation / relator	$u = v$ / the word uv^{-1}
word, reduced word	string over $S \cup S^{-1}$; no adjacent xx^{-1}	normal closure	smallest normal subgroup containing R
free group $F(S)$	reduced words, concatenate then reduce	Cayley graph	one vertex per element, one arrow per generator